



Common Symbols used in Set Theory

In the examples $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$

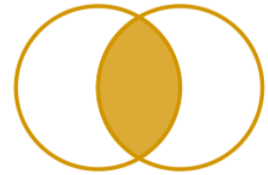
Symbol	Meaning	Example
$\{ \}$	Set A collection of elements	$\{1, 2, 3\}$
$A \cup B$	Union In A or B (or both)	$A \cup B = \{1, 2, 3, 4, 5\}$
$A \cap B$	Intersection In both A and B	$A \cap B = \{3, 4\}$
$A \subseteq B$	Subset Every element of A is in B	$\{3, 4, 5\} \subseteq B$
$A \subset B$	Proper Subset Every element of A is in B, but B has more elements	$\{3, 5\} \subset B$
$A \not\subset B$	Not a Subset A is not a subset of B	$\{1, 6\} \not\subset A$
$A \supseteq B$	Superset A has same elements as B, or more	$\{1, 2, 3\} \supseteq \{1, 2, 3\}$
$A \supset B$	Proper Superset A has B's elements and more	$\{1, 2, 3, 4\} \supset \{1, 2, 3\}$
$A \not\supset B$	Not a Superset A is not a superset of B	$\{1, 2, 6\} \not\supset \{1, 9\}$
A^c	Complement Elements not in A	$B^c = \{1, 2, 6, 7\}$ When $U = \{1, 2, 3, 4, 5, 6, 7\}$
$A - B$	Difference in A but not in B	$\{1, 2, 3, 4\} - \{3, 4\} = \{1, 2\}$
$a \in A$	Element of a is in A	$3 \in \{1, 2, 3, 4\}$
$b \notin A$	Not Element of b is not in A	$6 \notin \{1, 2, 3, 4\}$
$\emptyset$	Empty Set $= \{ \}$	$\{1, 2\} \cap \{3, 4\} = \emptyset$
U	Universal Set Set of all possible values (in the area of interest)	
$P(A)$	Power Set all subsets of A	$P\{1, 2\} = \{ \{ \}, \{1\}, \{2\}, \{1, 2\} \}$
$A = B$	Equality Both sets have the same members	$\{3, 4, 5\} = \{3, 4, 5\}$
$A \times B$	Cartesian Product (set of ordered pairs from A and B)	$\{1, 2\} \times \{3, 4\} = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$
$ A $	Cardinality The number of elements of set A	$ \{3, 4\} = 2$
$ $	Such that	$\{n \mid n > 0\}$ or $\{0, 1, 2, 3, \dots\}$
$:$	Such that	$\{n : n > 0\}$ or $\{0, 1, 2, 3, \dots\}$
$\forall$	For all	$\forall x > 1, x^2 > x$
$\exists$	There Exist	$\exists x \mid x^2 > x$
$\therefore$	Therefore	$a = b \therefore b = a$
$\mathbb{N}$	Natural Numbers	$\{1, 2, 3, \dots\}$ or $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	Integers	$\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
$\mathbb{Q}$	Rational Numbers	
$\mathbb{A}$	Algebraic Numbers	
$\mathbb{R}$	Real Numbers	
$\mathbb{I}$	Imaginary Numbers	$2i$
$\mathbb{C}$	Complex Numbers	$3 + 4i$

OPERATION OF SETS

Intersection

The intersection of two sets A and B, written $A \cap B$, is the set of elements that are in A and in B. In symbolic logic, a similar symbol, $\wedge$, is used to connect two logical propositions with the AND operator.

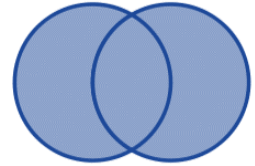
Excel Review Center



Intersection of Two Sets

Union

The union of two sets A and B, written $A \cup B$, is the set of elements that are in A or in B (or both). In logic a similar symbol, $\vee$, is used to connect two propositions with the OR operator.



Union of Two Sets

Difference

The difference of two sets A and B (also known as the set-theoretic difference of A and B, or the relative complement of B in A) is the set of elements that are in A but not in B. This is written $A - B$, or sometimes $A \setminus B$.

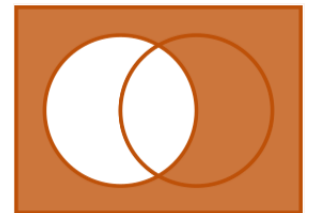
Difference ($A - B$)

Complement

The set of elements that are not in a set A is called the complement of A. It is written A' (or sometimes A^c , or $\bar{A}$).



Relative complement of A (left) in B (right)

Absolute complement of A in U
Excel Review Center

Cardinality

The cardinality of a finite set A, written $|A|$, is the number of (distinct) elements in A.

Excel Review Center

Power Sets

The power set of a set A is the set of all its subsets including itself and the empty set. It is denoted by $P(A)$.

Using set comprehension notation, $P(A)$ can be defined as:

$$P(A) = \{Q \mid Q \subseteq A\}$$

Example:

$$A = \{1, 2\}$$

$$P(A) = \{\{1, 2\}, \{1\}, \{2\}, \emptyset\}$$

Cardinality of Power Sets: If $|A| = k$, then $|P(A)| = 2^k$

Relation and Function

A **relation** is any association or link between elements of one set, called the **domain** or (less formally) the set of inputs, and another set, called the **range** or set of outputs.

A **function** is a relation that has exactly one output for every possible input in the domain.

Excel Review Center

Cartesian Products

Excel Review Center

Let A and B be sets. The Cartesian product of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

A subset of the Cartesian product $A \times B$ is called a **relation R** from the set A to the set B. The elements of R are ordered pairs, where the first element belongs to A and the second to B. The Cartesian products $A \times B$ and $B \times A$ are not equal.